ARJUNA JEE 2027

Lecture -02 | Mathematics

BINOMIAL THEOREM



By- Sachin Jakhar Sir



Topics *to be covered*

A **Middle Term**

B **Different Types of Questions**

C **Question Practice**

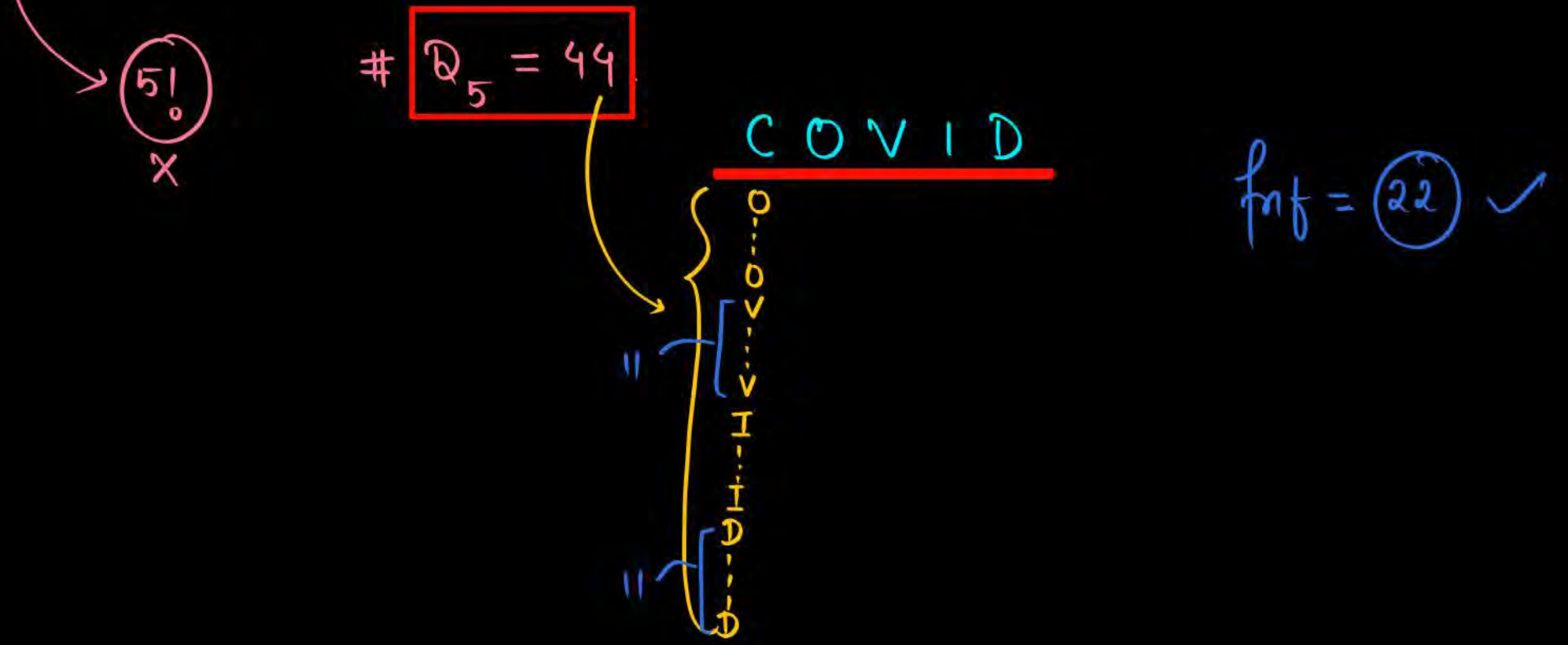
D



H.W. Discussion

Acidic Question

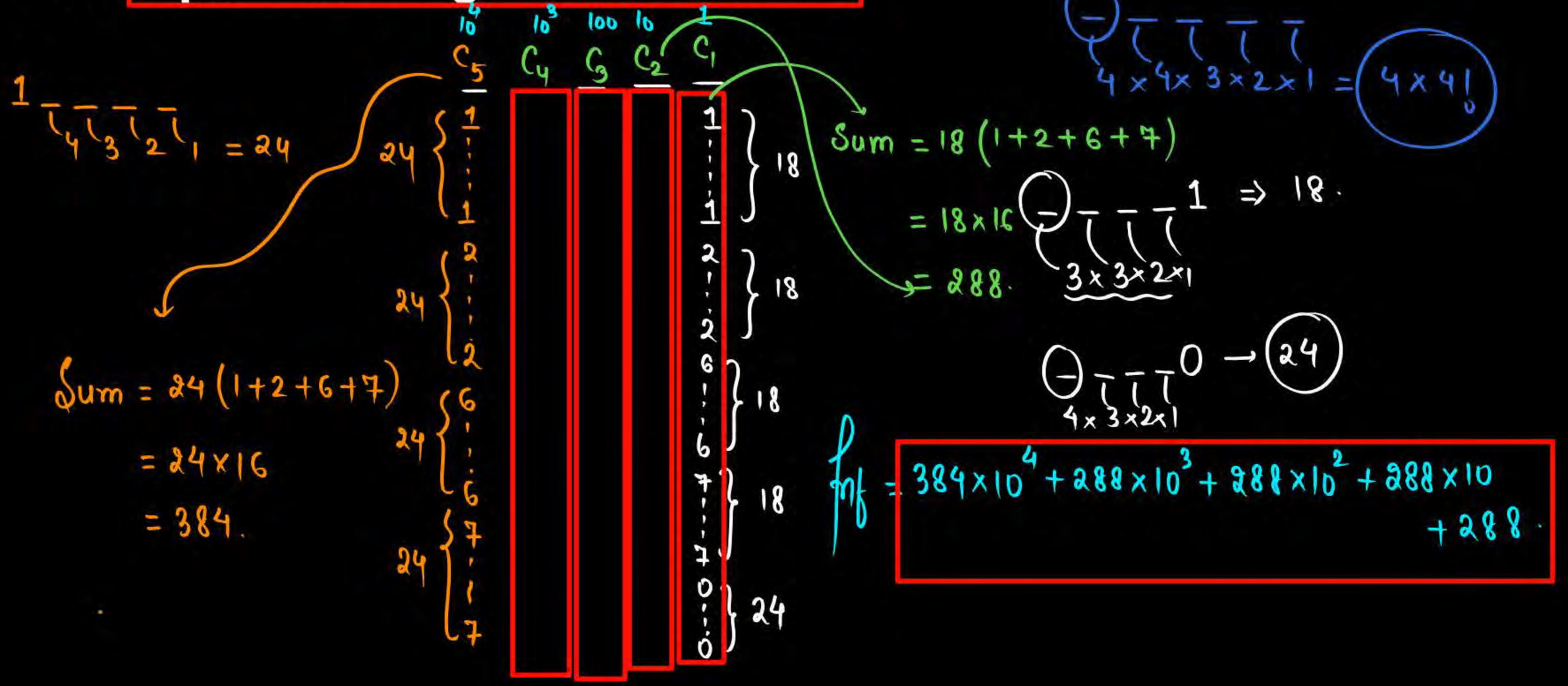
The number of words that can be formed by rearranging the letters of the word 'COVID' such that no letter is in it's own place and word should not start with a vowel is _____.



final = 22 ✓

Acidic Question

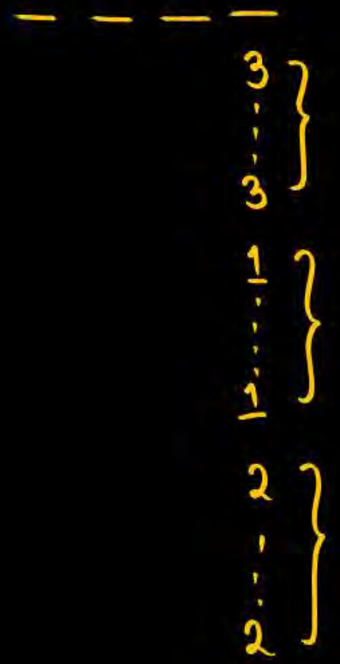
Find sum of all 5 digit numbers that can be formed with help of 0, 1, 2, 6, 7 in which repetition of digits is not allowed.



Question [JEE Mains-2023]

Sum of all the four digits numbers that can be formed using all the digits 2, 1, 2, 3 is_____.

$\frac{4!}{2!} = 12$



$f_{mf} = 26664$



EXPONENT OF PRIME NUMBER IN n!

Exponent of r in $n!$ = $\left[\frac{n}{r}\right] + \left[\frac{n}{r^2}\right] + \left[\frac{n}{r^3}\right] + \left[\frac{n}{r^4}\right] + \ldots.$

(where : $r \Rightarrow$ prime number, [.] $\Rightarrow$ Greatest Integer Function)

Also Note : $(2n!) = 2^n \cdot n!\,(1 \cdot 3 \cdot 5 \ldots\ldots\ldots (2n-1))$

$12! = 2^{10} \cdot 3^{5} \cdot 5^{2} \cdot 7^{1} \cdot 11^{1}$

$12! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 \cdot 11 \cdot 12$

$2^1 \quad 2^2 \quad 2^1 \cdot 3^1 \quad 2^3 \quad 3^2 \quad 2^1 \cdot 5^1 \quad 2^2 \cdot 3^1$

exponent of 2 in $12! = \left[\frac{12}{2}\right] + \left[\frac{12}{2^2}\right] + \left[\frac{12}{2^3}\right] + \left[\frac{12}{2^4}\right] + \cdots$

$= 6 + 3 + 1 + 0 + 0 + \cdots$

$= 10$

$$12!_0 = 1.\underline{2}.3.\underline{4}.5.\underline{6}.7.\underline{8}.9.\underline{10}.11.\underline{12}$$

$$= (1.3.5.7.9.11) \times (2.4.6.8.10.12)$$

2×1 2×2 2×3 2×4 2×5 2×6

$$(2\times6)!_0 = 12!_0 = 2^6 \cdot (1\times2\times3\times4\times5\times6) \times (1.3.5.7.9.\underbrace{11}_{2n-1})$$

$6!_0$

$$(2n)!_0 = 2^n \cdot n!_0 \cdot (1.3.5.\ldots.(2n-1))$$

$\alpha =$ _____
$\beta =$ _____

EXAMPLE

If $53! = 2^{\alpha}\,3^{\beta}\,5^{\gamma}\,7^{\delta}$ ....., then find γ = _____.

$$\gamma = \left[\frac{53}{5}\right] + \left[\frac{53}{5^2}\right] + \left[\frac{53}{5^3}\right] + \dots$$
$$= 10 + 2 + 0 = 12$$

$10 = 2 \times 5$
amount = 12.
amount = 49.

12 = exponent of 10 in $53!_0$ $= \left[\frac{53}{10}\right] + \left[\frac{53}{10^2}\right] + \dots = 5$ ✗ (because 10 is not a Prime No.)

" " 2 " " $= \left[\frac{53}{2}\right] + \left[\frac{53}{2^2}\right] + \left[\frac{53}{2^3}\right] + \left[\frac{53}{2^4}\right] + \left[\frac{53}{2^5}\right] + \left[\frac{53}{2^6}\right] + \dots$
$= 26 + 13 + 6 + 3 + 1 + 0$

$\alpha =$ _____
$\beta =$ _____

EXAMPLE

If $53! = 2^{\alpha}\, 3^{\beta}\, 5^{\gamma}\, 7^{\delta}$ ....., then find $\gamma =$ _____.

$\gamma = 12$

$$\gamma = \left[\frac{53}{5}\right] + \left[\frac{53}{5^2}\right] + \left[\frac{53}{5^3}\right] + \dots$$

$$= 10 + 2 + 0 = 12$$

$10 = 2 \times 5$

amount = 12.

amount = 49.

12 = exponent of 10 in $53!_0$ = no. of syphers in end of $53!_0$ = 12

zeroes.

Example

For 69!, find :

(i) Exponent of 2

(ii) Largest value n such that 69! is divisible by 3^n.

(iii) Exponent of 6 in 69!

(iv) Number of syphers in end of 69!



BINOMIAL THEOREM

$T_1 \quad T_2 \quad T_3 \qquad\qquad T_{n+1}$

$$\# \quad (a+b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \cdots\cdots + {}^nC_n a^0 b^n$$

$(a+b)$ ↑ = 2 terms

T.T. $= (n+1)$

$$\text{G.T.} \Rightarrow \boxed{T_{r+1} = {}^nC_r a^{n-r} b^r}$$

$$(a+b)^n = \sum_{r=0}^{n} {}^nC_r a^{n-r} b^r$$

ex:-

$(2x+3y)^{23}$

$n = 23$

Total = 24.

First = $2x$

Second = $3y$

$$T_{r+1} = {}^{23}C_r (2x)^{23-r} (3y)^r$$

7th Term $\Rightarrow$ $T_7 = T_{6+1} = {}^{23}C_6 (2x)^{23-6} (3y)^6$

(BC: ${}^{23}C_6$; 17: $23-6$; r: 6)

Coeff. of $T_7 = {}^{23}C_6 \, 2^{17} 3^6$

ex:-

$(2x+3y)^{23}$

$n = 23$

Total = 24.

First = $2x$

Second = $3y$.

k^{th} Term from last $= \left(\overbrace{\text{Total Term}}^{n+1} - k + 1\right)^{th}$ term from Starting.

11th Term from last $= (24 - 11 + 1)^{th}$ term from Starting

$= 14^{th}$ term from Starting

$T_{14} = T_{13+1} = {}^{23}C_{13}(2x)^{10}(3y)^{13} = \boxed{{}^{23}C_{13}}\, 2^{10}3^{13}x^{10}y^{13}$.

OR

11th Term from last of $(2x+3y)^{23} = 11^{th}$ Term from starting of $(3y+2x)^{23}$

$T_{11} = T_{10+1} = \boxed{{}^{23}C_{10}}(3y)^{13}\cdot(2x)^{10} = {}^{23}C_{10}\, 3^{13}2^{10}y^{13}x^{10}$

($\hookrightarrow r = 10$)



IMPORTANT NOTE

Power of binomial expression.

(v) k^{th} term from last = $(n - k + 2)^{th}$ term from starting.

(vi) Difference between coefficient and binomial coefficient.

(vii) To find "sum of all coefficients" put all parameters equal to 1.

Question [JEE Mains-2023]

The coefficient of x^5 in the expansion of $\left(2x^3 - \frac{1}{3x^2}\right)^5$ is

A $\frac{26}{3}$

B $\frac{80}{9}$

C 9

D 8

$$T_{r+1} = {}^5C_r \left(2x^3\right)^{5-r} \left(\frac{-1}{3x^2}\right)^r$$

$$= {}^5C_r \cdot \frac{2^{5-r} \cdot (-1)^r}{3^r} \cdot \frac{\left(x^3\right)^{5-r}}{\left(x^2\right)^r}$$

$$x^{15-3r-2r} = x^{15-5r}$$

$$T_{r+1} = {}^5C_r \frac{2^{5-r} \cdot (-1)^r}{3^r} x^{15-5r}$$

$15 - 5r = 5$

$10 = 5r$

$2 = r$

$r = 2$

$$T_3 = \frac{{}^5C_2 \cdot 2^3 \cdot 1}{3^2} x^5$$

$$G.T. = T_{r+1} = {}^{5}C_r \cdot \frac{2^{5-r} \cdot (-1)^r}{3^r} x^{15-5r}$$

$$\left(2x^3 - \frac{1}{3x^2}\right)^5$$

${}^{n}C_r$

$0 \le r \le n$

$r \in W$

$(1+x)^2 = x^2 + 2x + 1 + 0x^3$

coeff of x^3 in $(1+x)^2 = 0$.

coeff. of x^8 = zero.

$15 - 5r = 8$

$7 = 5r$

$\frac{7}{5} = r$ — n.p.

Question [JEE Mains-2021]

If the fourth term in the expansion of $\left(x + x^{\log_2 x}\right)^7$ is 4480, then the value of x where $x \in N$ is equal to

A 2

B 4

C 3

D 1

$T_4 = T_{3+1} = 4480$

$r = 3$

$${}^7C_3 \, x^4 \left(x^{\log_2 x}\right)^3 = 4480$$

$35 \times x^4 \left(x^{\log_2 x}\right)^3 = 35 \times 2^4 \times 2^3$

4×1120

4×280

$7 \times 4 \times 10 = 35 \times 8$

Question [JEE Mains-2013/21]

The term independent of x in expansion of $\left(\frac{x+1}{x^{2/3}-x^{1/3}+1}-\frac{x-1}{x-x^{1/2}}\right)^{10}$ is

A 4

B 120

C 210

D 310

a.

$$a=\frac{\left(x^{\frac{1}{3}}\right)^3+1^3}{\left(x^{\frac{1}{3}}\right)^2-\left(x^{\frac{1}{3}}\right)+1}=\left(x^{\frac{1}{3}}+1\right)\left(x^{\frac{2}{3}}+1-x^{\frac{1}{3}}\right)$$

$$a=x^{\frac{1}{3}}+1.$$

$$\frac{x-1}{x-\sqrt{x}}=\frac{(\sqrt{x}-1)(\sqrt{x}+1)}{\sqrt{x}(\sqrt{x}-1)}=\frac{\sqrt{x}+1}{\sqrt{x}}=1+\frac{1}{\sqrt{x}}.$$

$(\sqrt{x})^2-1^2$

$$\left(x^{\frac{1}{3}}+x-\left(x+\frac{1}{\sqrt{x}}\right)\right)^{10}$$

$$\left(x^{\frac{1}{3}}-\frac{1}{\sqrt{x}}\right)^{10}.$$

$$\left(x^{\frac{1}{3}} - \frac{1}{\sqrt{x}}\right)^{10}$$

$$T_{r+1} = {}^{10}C_r \left(x^{\frac{1}{3}}\right)^{10-r} \left(\frac{-1}{\sqrt{x}}\right)^r$$

$$= {}^{10}C_r \cdot x^{\frac{10-r}{3}} \frac{(-1)^r}{x^{r/2}} = {}^{10}C_r \cdot (-1)^r \, x^{\left(\frac{10-r}{3} - \frac{r}{2}\right)}$$

T_5

$$\left(\frac{10-r}{3} - \frac{r}{2}\right) = 0$$

$$\frac{10-r}{3} = \frac{r}{2}$$

$$20 - 2r = 3r$$

$$20 = 5r$$

$$r = 4$$

$$T_5 = {}^{10}C_4 = \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} = 210$$

Question [JEE Mains-2024]

If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal

A 6

B 4

C 2

D 9

90% DIBY
HW

Question [JEE (Adv.)-2023

[Ans. : 3]

Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2+\frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax-\frac{1}{bx^2}\right)^7$, then the value of $2b$ is



Question [JEE (Adv.)-2013 & Mains-2020]

Coefficient of 3 consecutive terms in expansion of $(1 + x)^{n+5}$ are in ratio $5 : 10 : 14$ then n =_____.

T_r T_{r+1} T_{r+2}

$^{n+5}C_{r-1}\, x^{r-1}$ $^{n+5}C_{r}\, x^{r}$ $^{n+5}C_{r+1}\, x^{r+1}$

$$^{n+5}C_{r-1} : {}^{n+5}C_{r} : {}^{n+5}C_{r+1} = 5 : 10 : 14.$$

Question [JEE Mains-2019]

Coefficient of 3 consecutive terms in expansion of $(x + 1)^n$ are in ratio $2 : 15 : 70$ then average of all is _232___.

DIBY !!

$${}^{n+5}C_{r-1} : {}^{n+5}C_{r} : {}^{n+5}C_{r+1} = 5 : 10 : 14.$$

$$\frac{{}^{n+5}C_{r-1}}{{}^{n+5}C_{r}} = \frac{1}{2}$$

$$\frac{{}^{n+5}C_{r}}{{}^{n+5}C_{r+1}} = \frac{10}{14} = \frac{5}{7}$$

$$\frac{(n+5)!\; r!\; (n-r+5)!}{(r-1)!\; (n-r+6)!\; (n+5)!} = \frac{1}{2}$$

$$\frac{(n+5)!\; (r+1)!\; (n-r+4)!}{r!\; (n-r+5)!\; (n+5)!} = \frac{5}{7}$$

$$(n-r+6)\cdot(n-r+5)!$$

$$\frac{r}{n-r+6} = \frac{1}{2}$$

$$\frac{r+1}{n-r+5} = \frac{5}{7}$$

$$n = 6$$

Question [JEE Mains-2024]

Let m and n be the coefficients of seventh and thirteenth terms respectively in the expansion of $\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$. Then $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ is :



A $\frac{4}{9}$

B $\frac{1}{9}$

C $\frac{1}{4}$

D $\frac{9}{4}$

If the sum of the coefficients of all the positive even powers of x in the binomial expansion of $\left(2x^3 + \frac{3}{x}\right)^{10}$ is $5^{10} - \beta \cdot 3^9$, then β is equal to __83__.

$$T_{r+1} = {}^{10}C_r (2x^3)^{10-r} \left(\frac{3}{x}\right)^r$$

$$= {}^{10}C_r \cdot 2^{10-r} 3^r \cdot x^{30-4r}$$

${}^{10}C_r \Rightarrow r = 0, 1, 2, 3 \ldots 10$

	$30-4r$
$r=0$	30
$r=1$	26
$r=2$	22
$r=3$	18
⋮	
$r=7$	2

$r=8$	-2
$r=9$	-6
$r=10$	-10

Sum of all the coeff. ($x=1$) $= 5^{10}$.

$$\text{Pf} = 5^{10} - \left({}^{10}C_8 2^2 \cdot 3^8 + {}^{10}C_9 2^1 \cdot 3^9 + {}^{10}C_{10} 3^{10} \right)$$

$$= 5^{10} - 3^9 \left(\frac{{}^{10}C_8 \cdot 2^2}{3} + {}^{10}C_9 \cdot 2^1 + {}^{10}C_{10} \cdot 3^1 \right)$$

β

$$(a+b)^n = \sum_{r=0}^{n} {}^nC_r\, a^{n-r} b^r$$

$$\sum_{r=0}^{5} {}^5C_r\, \underline{2}^{5-r}\, \underline{3}^{r} = (2+3)^5 = 5^5$$

$$\sum_{r=0}^{5} {}^5C_r \cdot \underline{(1)}^{5-r}\, \underline{3}^{r} = (1+3)^5 = 4^5$$

$$\sum_{r=0}^{5} {}^5C_r\, (-1)^r \cdot 4^{5-r} = \sum_{r=0}^{\textcircled{5}} \textcircled{5}C_r\, \underline{4}^{5-r}\, (-1)^r \cdot = (4+(-1))^5 = 3^5$$

Question [JEE Mains-2021]

Let n be a positive integer. Let

$$A = \sum_{k=0}^{n} (-1)^k {}^nC_k \left[\left(\frac{1}{2}\right)^k + \left(\frac{3}{4}\right)^k + \left(\frac{7}{8}\right)^k + \left(\frac{15}{16}\right)^k + \left(\frac{31}{32}\right)^k\right]$$. If $63A = 1 - \frac{1}{2^{30}}$, then n is equal to

$n = 6$ fnf.

$$\sum_{r=0}^{n} {}^nC_r 1\left(-\frac{1}{2}\right)^r = \left(1 - \frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^n$$

$$\sum_{r=0}^{n} {}^nC_r 1\left(-\frac{3}{4}\right)^r = \left(1 - \frac{3}{4}\right)^n = \left(\frac{1}{4}\right)^n$$

$$\sum_{r=0}^{n} {}^nC_r 1\left(-\frac{7}{8}\right)^r = \left(1 - \frac{7}{8}\right)^n = \left(\frac{1}{8}\right)^n$$

$$A = \left(\frac{1}{2}\right)^n + \left(\frac{1}{2}\right)^{2n} + \left(\frac{1}{2}\right)^{3n} + \left(\frac{1}{2}\right)^{4n} + \left(\frac{1}{2}\right)^{5n}$$

G.P.

$$A = \left(\frac{1}{2}\right)^n \left(\frac{1 - \left(\left(\frac{1}{2}\right)^n\right)^5}{1 - \left(\frac{1}{2}\right)^n}\right) = \frac{1}{2^n}\left(\frac{2^{5n} - 1}{\frac{2^{5n}}{2^n - 1}}\right)$$

$$A = \frac{2^{5n} - 1}{2^{5n}\left(2^n - 1\right)} \Rightarrow \left(2^n - 1\right)A = 1 - \frac{1}{2^{5n}}$$

• Middle Term

$(a+b)^n$

$T.T. = (n+1)$

$\frac{n+1}{2}, \frac{n+2}{2}, \frac{n+3}{2}$

Q+S

$a_1\ a_2\ a_3\ \underset{\uparrow}{a_4}\ a_5\ a_6\ a_7$ $\quad T.T. = 7$

$a_1\ a_2\ a_3\ \underset{\uparrow}{a_4}\ \underset{\uparrow}{a_5}\ a_6\ a_7\ a_8$ $\quad T.T. = 8$

$\frac{TT}{2}, \frac{TT+1}{2}, \frac{TT+2}{2}$

$\frac{n}{2}, \frac{n+1}{2}, \frac{n+2}{2}$

$\left(\frac{2}{x} - 3y\right)^{23} = n$

Middle Term → T_{12} & T_{13} ✓

$\frac{23+1}{2}$, $\frac{23+2}{2}$, $\frac{23+3}{2}$

12 , 13

$T_{11+1} = {}^{23}C_{11}\left(\frac{2}{x}\right)^{12}\cdot(-3y)^{11}$

MIDDLE TERM

Calculate : $\frac{n+1}{2}, \frac{n+2}{2}, \frac{n+3}{2}$

$\Rightarrow$ If $n \in$ even $\Rightarrow$ Middle term = $\left(\frac{n+2}{2}\right)^{\text{th}}$ terms

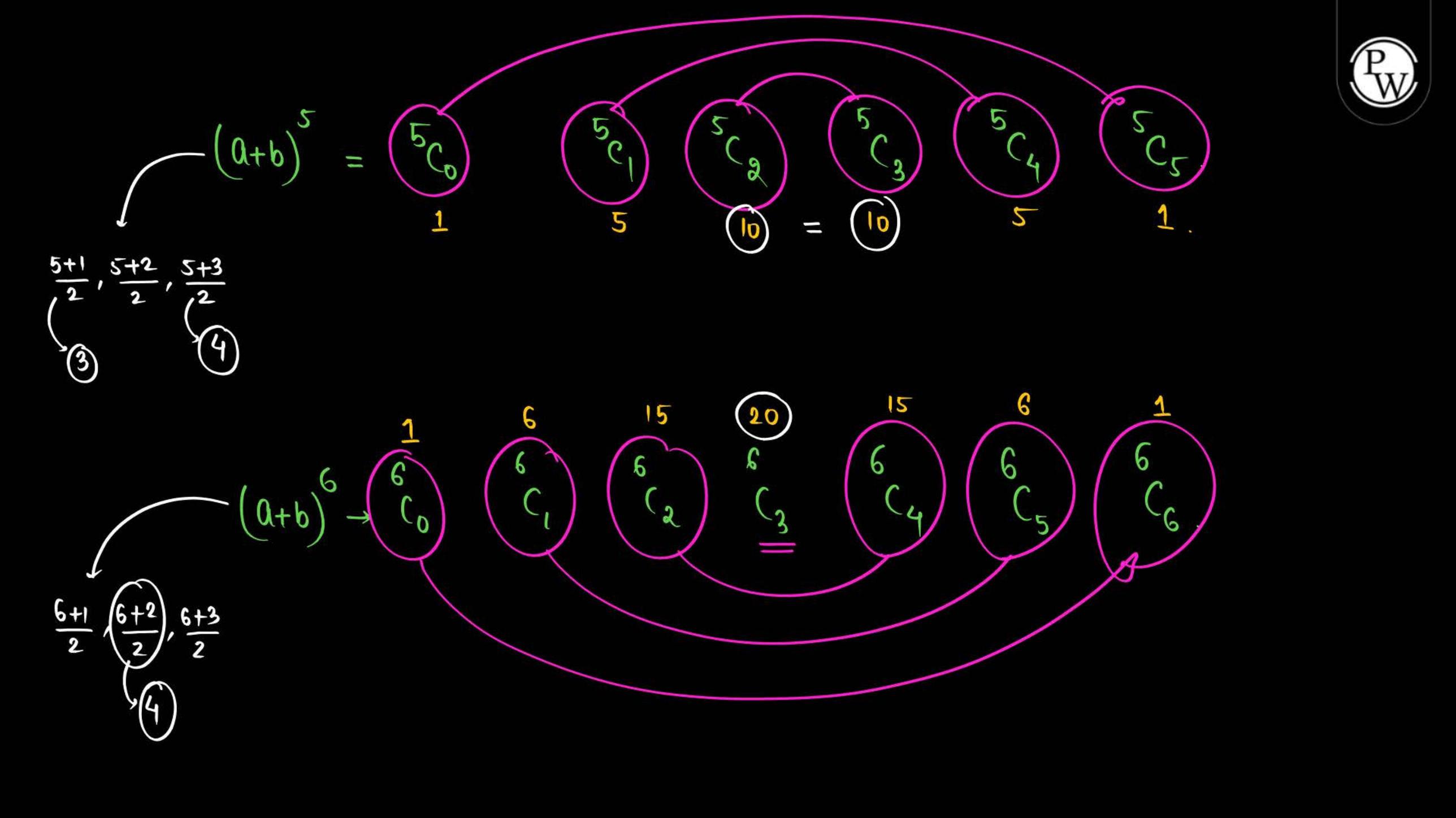
$\Rightarrow$ If $n \in$ odd $\Rightarrow$ Middle term = $\left(\frac{n+1}{2}\right)^{\text{th}}$ and $\left(\frac{n+3}{2}\right)^{\text{th}}$ term } ignore !!

Note:

(1) Maximum value of ${}^{n}C_{r}$ is when $r = \frac{n}{2}, n \in$ even

$r = \frac{n+1}{2}$ or $\frac{n-1}{2}, n \in$ odd

(2) Middle term has largest binomial coefficient.


$(a+b)^5 = {}^5C_0 \quad {}^5C_1 \quad {}^5C_2 \quad {}^5C_3 \quad {}^5C_4 \quad {}^5C_5$
1 5 10 = 10 5 1.
$\frac{5+1}{2}, \frac{5+2}{2}, \frac{5+3}{2}$
3 4
$(a+b)^6 \rightarrow {}^6C_0 \quad {}^6C_1 \quad {}^6C_2 \quad {}^6C_3 \quad {}^6C_4 \quad {}^6C_5 \quad {}^6C_6$
1 6 15 20 15 6 1
$\frac{6+1}{2}, \frac{6+2}{2}, \frac{6+3}{2}$
4

$${}^{n}C_{r}\Big|_{\max.} \Rightarrow r = \frac{n}{2}$$

$$r = \frac{n+1}{2} \text{ OR } \frac{n-1}{2}$$

$\overset{\equiv}{(12)}C_{r}\Big|_{\max.} \Rightarrow r = \frac{12}{2} = 6$

$\equiv$

$${}^{12}C_{6}$$

$\equiv$

$(13)C_{r}$

$${}^{13}C_{7} = {}^{13}C_{6}$$

max.

$$r = \frac{13+1}{2}, \frac{13-1}{2}$$

$$r = 7, 6$$

DO IT BY YOURSELF (DIBY)

DIBY-01

The coefficient of x^{-6} in the expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is _____ [JEE Mains-2023]

[Ans. : 5040]

DIBY-02

The natural number m, for which the coefficient of x in the binomial expansion of $\left(x^m+\frac{1}{x^2}\right)^{22}$ is 1540, is _____ [JEE Mains-2020]

[Ans. : 13]

DIBY-03

The sum of the real values of x for which the middle term in the binomial expansion of $\left(\frac{x^3}{3}+\frac{3}{x}\right)^8$ equals 5670 is [JEE Mains-2019]

(A) 0 (B) 8 (C) 6 (D) 4

[Ans. : (A)]

DIBY-04

The ratio of the coefficient of the middle term in the expansion of $(1 + x)^{20}$ and the sum of the coefficients of two middle terms in the expansion of $(1 + x)^{19}$ is

[JEE Mains-2021]

[Ans. : 1]

DIBY-05

If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10} =$ is 180, then r is equal to

[JEE Mains-2021]

[Ans. : ....]

DIBY-06

If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal to

[JEE Mains-2024]

(A) 2 (B) 4 (C) 6 (D) 9

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians